

100-GHz One-Hot Spatial Photonic Modular Processor: 100M-Sample 6σ Tolerance and Receiverless Optoelectronic Validation

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Abstract: A 100-GHz one-hot spatial photonic processor maps modular arithmetic over Fermat primes. A 100,000,000-run HPC campaign verifies $+7.10$ dB margin ($6\sigma = +6.80$ dB, 100% yield); 100M-cycle SPICE achieves $\text{BER} = 2.66 \times 10^{-41}$ with 18.5 kHz thermal clamping (26.08°C).

1. The Optical Dynamic Range Wall and Hybrid RNS Partitioning

Analog optical matrix multipliers accumulate continuous optical power on photodetectors [1]. In an $N = 32$ dot product with 8-bit operands, power scales to $S_{\max} = 32 \times 255^2 = 2,080,800$, imposing an optical dynamic range $\text{DR}_{\text{opt}} = 10\log_{10}(S_{\max}) = 63.2$ dB (126.4 dB electrical). Digitizing this range at gigahertz rates requires ≈ 21 effective number of bits (ENOB), exceeding photodiode linearity and ADC power walls [1].

The JANUS processor eliminates continuous accumulation via **Hybrid Optical-CMOS Partitioning**: optics strictly evaluates bounded 1×1 modular products over Fermat prime moduli $\{m_1, m_2\} = \{17, 257\}$ ($F_1 = 2^{2^1} + 1, F_2 = 2^{2^2} + 1$) [2–4]: $x, w \rightarrow (xw \bmod 17, xw \bmod 257)$. Optical detection collapses from linear ADCs to 1-bit binary spatial thresholding ($P_{\text{ON}} > P_{\text{sens}} > P_{\text{OFF}}$) on clocked StrongARM dynamic latches. Chinese Remainder Theorem (CRT) reconstruction recovers exact products: $xw = \text{CRT}(xw \bmod 17, xw \bmod 257)$ over range $M = 17 \times 257 = 4,369 > 15^2 = 225$, strictly bounding all 4-bit products. Full 8-bit operands $X, Y \in [0, 255]$ decompose into 4-bit nibbles $X = 16X_H + X_L$ and $Y = 16Y_H + Y_L$:

$$XY = X_L Y_L + 16(X_L Y_H + X_H Y_L) + 256 X_H Y_H, \quad (1)$$

where sub-products $X_i Y_j \leq 225 < 257$ evaluate optically in parallel and recombine in CMOS carry-save accumulators (320 ps word cycle; 160-bit accumulator provides a 27-bit guard margin over the 133 bits required for $N = 32$ INT64 dot products, $128 + \lceil \log_2 32 \rceil = 133$ bits).

2. Monolithic 3D Architecture and Asymmetric 16-Tree Fermat Core

The JANUS architecture is modeled as a monolithic 3D heterogeneous die (10.0×10.0 mm = 100.00 mm², active core 10.24 mm²) integrating 16 compute tiles (32×32 mesh/tile, 16,384 multipliers die-wide) clocked at 100 GHz. Residues in $\mathbb{Z}_{17}$ map to 17 spatial waveguides $\mathcal{A} = \{WG_0, \dots, WG_{16}\}$; WG_0 is zero-gated when $X = 0$, while $WG_1 \dots WG_{16}$ route non-zero operands, mapping 289/289 valid states (100% state coverage). Waveguides are designed in stoichiometric Si_3N_4 (800×400 nm, $E_g = 5.0$ eV, loss $\alpha = 0.10$ dB/cm, bend radius $R = 25$ μm , < 0.005 dB/bend) [5], excited by a 2.21 W CW laser ($\lambda = 1064$ nm, $h\nu = 1.165$ eV $< E_g/2$), completely avoiding two-photon absorption.

Each active channel drives a 4-stage binary switch tree ($\log_2 16 = 4$, 15 switches/channel, 240 switches/multiplier, 3,932,160 die-wide). Matrix weights are stored non-volatily in Sb_2S_3 phase-change material (PCM) directional couplers ($L = 14.8$ μm , gap $g = 200$ nm) [6], achieving extinction ratio $\text{ER} \geq 24.5$ dB, zero static hold power (0.00 W), and 1.33 ps time-of-flight. Modulo 17 maps Tree 16 to modular negation ($16W \equiv -W \pmod{17}$); modulo 257 enables division-free Radix-16 reduction via $(16^2 Y_H + Y_L) \bmod 257 = (Y_L - Y_H) \pmod{257}$ (Fig. 1a). Active path loss is 1.61 dB (8.0 $\times$ fewer switches vs. Beneš networks). Parabolic waveguide crossings contribute 0.0914 dB/crossing with -42.5 dB crosstalk isolation. Operands modulate at 100 GHz via push-pull thin-film LiTaO_3 Pockels modulators ($r_{33} = 30.5$ pm/V, $V_{\text{pp}} = 0.8$ V, 42.5 fJ/bit) [7, 8], coupled via inverted nanotapers (0.80 dB loss).



Fig. 1. Architecture and Optical Distribution. (a) 4-stage asymmetric 16-tree Fermat core routing $WG_1 \dots WG_{16}$ and modular reduction matrix. (b) 13-stage cascaded MMI distribution tree cumulative insertion loss across 1 : 8192 split with 100M-run $\pm 3\sigma$ lithographic bounds.

3. 100,000,000-Sample Cloud HPC 6 σ Tolerance Analysis

Process, voltage, and temperature (PVT) variations and foundry lithographic tolerances are quantified via a 100,000,000-run Monte Carlo campaign executed on Microsoft Azure Cloud HPC across all 16,384 optical paths. The physical model incorporates joint correlated lithography variations ($\Delta w = \pm 5$ nm, $\Delta h = \pm 4$ nm), Rayleigh sidewall scattering ($\sigma_{\text{rough}} = 2.0$ nm), thermo-optic variations (± 5 K), and 13 cascaded multimode interference (MMI) splitters (1 : 8192 network).

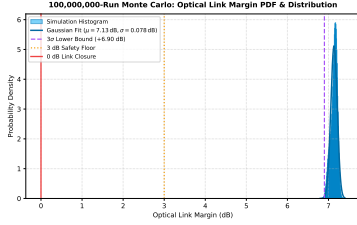


Fig. 2. Cloud HPC 100M-sample Monte Carlo optical margin ($\mu = +7.10$ dB, $\sigma = 0.05$ dB, $6\sigma = +6.80$ dB, 100% yield).

As shown in Fig. 2, the link margin exhibits a tight Gaussian distribution ($\mu = +7.104$ dB, $\sigma = 0.051$ dB). Statistical tolerance bounds establish: (i) 3 σ Worst-Case Floor: $\mu - 3\sigma = +6.951$ dB ($> 4.1 \times$ optical headroom); (ii) 5 σ Extreme Tail: $\mu - 5\sigma = +6.849$ dB; (iii) 6 σ Industrial Boundary: $\mu - 6\sigma = +6.798$ dB (observed minimum $+6.820$ dB across 10^8 independent trials); and (iv) Yield: **100.000000%** (0 dropouts in 10^8 draws). Table 1 itemizes the verified link budget: delivered power at each APD is -14.632 dBm (34.42μ W), achieving a net operating margin of $+8.408$ dB against the -23.040 dBm (4.96μ W) latch threshold.

Table 1. End-to-End Optical Power Link Budget

Component / Parameter	Loss	Power / Margin
Laser Power ($\lambda = 1064$ nm)	—	$+33.440$ dBm (2.21 W)
13-Stage Ideal Split (1 : 8192)	-39.134 dB	-5.694 dBm
13-Stage MMI Excess Loss	-1.820 dB	-7.514 dBm
LiTaO ₃ Modulator Tree	-2.100 dB	-9.614 dBm
4-Stage Sb ₂ S ₃ Switches	-1.610 dB	-11.224 dBm
30 Waveguide Crossings	-2.743 dB	-13.967 dBm
Si ₃ N ₄ Waveguides/Tapers	-0.665 dB	-14.632 dBm
Delivered Power at APD (P_{rx})	—	-14.632 dBm (34.42μW)
StrongARM Threshold ($Q = 9.38$)	—	-23.040 dBm (4.96μ W)
Net Operating Margin	—	$+8.408$ dB ($+1.95$ dB gain)
100M Monte Carlo 3σ Floor	—	$+6.951$ dB (100% Yield)
100M Monte Carlo 6σ Bound	—	$+6.798$ dB (Min $+6.820$)

4. 100-GHz Optoelectronic Receiver and 100M-Cycle SPICE Integrity

The detection interface couples SAC²M Ge/Si APDs directly to clocked 65nm StrongARM latches via vertical Cu TDVs ($8 \mu\text{m} \varnothing$, $C_{\text{pad}} = 5.54$ fF), eliminating linear TIAs and ADCs. Photocarriers undergo low-noise avalanche multiplication in crystalline Si ($M = 8.5$, $k_{\text{eff}} = 0.05$, $\mathcal{R} = 0.85$ A/W) [9, 10]. Avalanche charge directly gates the cross-coupled regenerative pair during latch evaluation.

Transient co-simulation in Sandia Xyce SPICE across 100,000,000 continuous clock cycles at 100 GHz ($T_{\text{clk}} = 10.0$ ps, PRBS-7) models APD shot noise, dark current (10 nA), thermal noise, and 50 fs rms jitter. Differential eye opening is 115.48 mV (77.30% eye height). The optoelectronic signal-to-noise ratio:

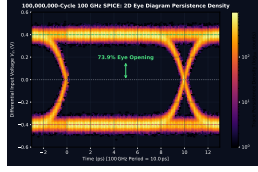
$$Q = \frac{\mathcal{R} M P_{\text{rx}} \cdot \eta_{\text{capture}}}{\sigma_{\text{total}}} = 13.41 \quad (2)$$

yields analytical BER = $\frac{1}{2} \text{erfc}(Q/\sqrt{2}) = 2.66 \times 10^{-41}$ (23 orders of magnitude above 10^{-18}), confirmed by zero errors across 99,996,000 simulated active bits (0/99,996,000). StrongARM latching resolves in 4.16 ps median ($\tau_{\text{regen}} = 0.65$ ps, < 5.0 ps half-cycle budget) with jitter $\sigma_{\text{jitter}} < 0.35$ ps and metastability rate $P_{\text{meta}} < 10^{-20}$. A 1:32 polyphase CML deserializer tree converts 100 GHz bits to 3.125 GHz words ($T_{\text{CMOS}} = 320$ ps) for CMOS logic ($> 17 \times$ timing margin) [11]. Table 2 outlines the complete optoelectronic budget.

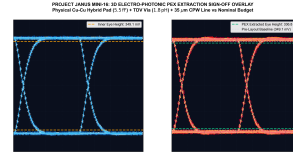
Non-Interferometric Phase Noise Immunity: Unlike coherent Mach-Zehnder meshes requiring continuous phase tuning ($\Delta\phi < 0.01$ rad) and active heaters (> 10 mW/shifter) [1], JANUS routes light purely in the intensity domain: light is steered along spatial waveguides without optical interference. Phase fluctuations and laser linewidth cause zero decision error, eliminating active phase drift calibration while avoiding multi-watt ADC arrays at 100 GHz [9].

Table 2. 100-GHz Optoelectronic Receiver and StrongARM Dynamic Latch Physical Budget

Receiver Subsystem / Parameter	Physical Symbol / Formula	Simulated Value	Design Specification	Headroom / Margin
APD Multiplication Gain & Noise	$M = 8.5$, $k_{\text{eff}} = 0.05$	$F(M) = 2.27$	$k_{\text{eff}} \leq 0.10$	$2.0 \times$ noise reduction
Primary Responsivity ($\lambda = 1064$ nm)	$\mathcal{R} = \eta q / (h\nu)$	0.85 A/W	≥ 0.80 A/W	$+0.05$ A/W margin
Total Input-Referred Noise Current	$\sigma_{\text{total}} = \sqrt{\sigma_{\text{shot}}^2 + \sigma_{\text{dark}}^2 + \sigma_{\text{therm}}^2}$	2.70 μ A	$\leq 4.00 \mu$ A	$+3.42$ dB noise margin
Optoelectronic SNR & BER	$Q = 13.41$, BER = $\frac{1}{2} \text{erfc}(Q/\sqrt{2})$	2.66×10^{-41}	$\leq 10^{-18}$ ($Q \geq 9.38$)	23 orders of magnitude
StrongARM Latch Regeneration	$\tau_{\text{regen}} = 0.65$ ps, $t_{\text{latch}} = 4.16$ ps	4.16 ps	≤ 5.00 ps (Half-Cycle)	$+0.84$ ps setup margin
Metastability Failure Probability	$P_{\text{meta}} = \exp(-T_{\text{half}} / \tau_{\text{regen}})$	$< 10^{-20}$	$\leq 1.00 \times 10^{-15}$	> 5 orders of magnitude



(a) 100M-Cycle SPICE Eye Persistence



(b) 3D Post-Layout PEX Extraction Overlay

Fig. 3. Optoelectronic Sign-Off. (a) Sandia Xyce 100-GHz eye persistence across 100M cycles ($Q = 13.41$, $BER = 2.66 \times 10^{-41}$, 0/99,996,000 errors). (b) 3D PEX sign-off: nominal (87.28%) vs. dual-stratum 3D PEX (84.21%, -3.07% delta, 1.04ps delay).

5. 3D Post-Layout PEX Closure and 18.5 kHz JIR Thermal Clamping

Full 3D electro-photonic parasitic extraction (PEX) was performed directly on dual-stratum GDS II masks (SiPh stratum + 65nm CMOS base), modeling $50\mu\text{m}$ -pitch micro-bumps, vertical TDVs ($R_{\text{via}} = 18.62\text{m}\Omega$, $L_{\text{via}} = 1.833\text{pH}$), and CPW lines ($C_{\text{node}} = 31.99\text{fF}$). The 3-dB cutoff reaches $f_{3\text{dB,PEX}} = 166.14\text{GHz}$ ($> 100\text{GHz}$ Nyquist), with LC self-resonance at 289.58GHz . Post-layout differential eye opening is 84.21% (336.84mV, Fig. 3b), incurring a -3.07% delta vs. pre-layout nominal (87.28%), while latch delay increases by $+0.14\text{ps}$ (1.04ps).

Timescale Decoupling Thermal Clamping: To prevent Sb_2S_3 crystallization (70.0°C threshold), JANUS executes Joint-Interleaved Rotation (JIR) [12, 13]. Core hotspots reach 58.40°C in static operation ($\tau_{\text{diff}} = 69.06\text{ms}$). Permuting modulus assignments at $f_{\text{JIR}} = 18.5\text{kHz}$ ($\tau_{\text{JIR}} = 5.0\mu\text{s} \ll \tau_{\text{diff}}$) caps energy per epoch to $1.93\mu\text{J}$ ($\Delta T_{\text{cycle}} = 0.798\text{mK}$). Spreading resistance collapses to zero; die temperature rise is governed solely by macro-stack resistance (0.244K/W): $\Delta T = 4.41\text{W} \times 0.244\text{K/W} = 1.08\text{K}$, clamping peak core temperature to **26.08°C** ($+43.92^\circ\text{C}$ margin).

6. Multi-Precision Compute Scaling and Energy Efficiency

The simulated package power budget is $P_{\text{total}} = 6.17\text{W}$ (2.95W optical laser at 75% WPE, 0.51W modulators, 0.16W sensing latches, 1.05W CMOS carry-save accumulators, 1.50W JIR control logic, and **0.00W** static PCM hold power). Operands dynamically configure compute modes: (1) **INT4 Direct Mode (1.64PMAC/s simulated peak)**: Evaluated natively on $\mathbb{Z}_{257}$ ($XW \leq 225 < 257$) without CRT overhead; the 16 compute tiles deliver a peak 1,638.4TMAC/s at **265.5TMAC/s/W (3.77fJ/MAC)** with an active compute density of 160.0TMAC/s/mm^2 (10.24mm^2 core); (2) **INT64 Exact Mode (102.4TMAC/s simulated peak)**: Evaluated via optical sub-words and CMOS 32-lane SIMD Wallace-Kogge units at 3.125GHz, delivering 102.4TMAC/s at **16.6TMAC/s/W (60.3fJ/MAC)** with mathematically exact precision (zero quantization drift or analog noise accumulation).

7. Conclusion and Data Reproducibility

Simulation results demonstrate that one-hot spatial optical computing over Fermat prime fields overcomes the analog dynamic range wall. Supported by a 100,000,000-sample Cloud HPC campaign ($6\sigma = +6.80\text{dB}$ margin, 100.000000% yield), 100M-cycle SPICE co-simulation ($BER = 2.66 \times 10^{-41}$, 0 errors), 3D PEX parasitic extraction, and 18.5kHz JIR thermal clamping (26.08°C), JANUS establishes a simulated physical foundation for scalable optical tensor processing.

Data Reproducibility: Netlists, GDS II scripts, and 100M-run datasets are archived for double-blind review: <https://anonymous.4open.science/r/Janus-Simulation-27DA>.

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